

GULF OF MEXICO BILGE DISCHARGE INCIDENT

Issuing Authority: Indian Coast Guard / DG Shipping | Investigator: Cmdr. Rajesh Sharma, Maritime Enforcement |
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1. SATELLITE DETECTION & DRIFT MODELING

Incident Ref / ID:	SPILL-SAR-9512	SAR Satellite Sensor:	Sentinel-1 SAR (C-band VV)
Detection Time:	2026-09-10T04:15:00Z	Detection Confidence:	74.1% (Physics-based)
Detected Centroid:	28.0057°N, -90.9021°E	Slick Surface Area:	400.85 km² (288.8 km perim)
Probable Origin Centroid:	27.9291°N, -91.0572°E	Est. Release Window:	2026-09-09T18:15 to 22:15 UTC
Drift Duration:	8.0 Hours Backward	Net Drift Vector:	1.2 kt @ 61°

2. RANKED CULPRIT VESSEL CORRELATION (NOAA MarineCadastre AIS)

Rank	Vessel Name	MMSI	Type	Closest Dist	Suspicion Score	Key Findings
#1	OCEAN TITAN	368123456	Tanker (Crude/Chem	0.0 km	95.4% [HIGH]	AIS Blackout (75m), Direct Origin Transit
#2	SEA PHANTOM	338987654	Cargo / Container	8.6 km	42.7% [LOW]	Spoofing Alert
#3	PACIFIC TRADER	413555888	Cargo / Container	17.8 km	29.8% [LOW]	Normal Passage
#4	GULF HARVESTER	367444111	Fishing Vessel	24.2 km	27.2% [LOW]	Normal Passage
#5	DELTA PIONEER	356889001	Tug / Towing / Pil	33.2 km	24.8% [LOW]	Normal Passage
#6	NORDIC HORIZON	219001234	Cargo / Container	16.2 km	12.3% [LOW]	Normal Passage

3. PRIMARY SUSPECT FORENSIC DOSSIER: OCEAN TITAN

Vessel Name:	OCEAN TITAN	MMSI / IMO:	368123456 / IMO9842109
Vessel Classification:	Tanker (Crude/Chemical/Gas)	Dimensions / Draft:	Length: 274.0 m Draft: 15.2 m
Closest Approach (CPA):	0.00 km to spill origin	CPA Timestamp:	2026-09-09T20:49:30+00:00

Scoring Formula:	Score Breakdown: Proximity (35.0%) + Temporal (23.9%) + AIS Gap (19.5%) + Anomaly (17.0%) = 95.4%	Dark Gap Blackout:	75 Minutes Silenced
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Forensic Evidence & Maneuver Observations:

- DARK TRANSIT: Vessel traversed directly through origin zone while AIS transponder was silenced (75 min blackout)
- Suspicious AIS blackout of 75 min during spill release window
- Sudden speed reduction: Cruising at 13.1 kt, decelerated to 4.2 kt near spill origin
- High-risk vessel class: Crude/Chemical Tanker subject to MARPOL Annex I Special Area regulations

LEGAL JURISDICTION & ENFORCEMENT NOTICE: This forensic evidence dossier has been generated by the SPILLX automated platform in accordance with International Maritime Organization (IMO) MARPOL 73/78 Annex I guidelines (Regulations for the Prevention of Pollution by Oil). The spatial-temporal correlation, Lagrangian drift origin estimation, and AIS dark-vessel blackout records constitute actionable evidentiary material for Port State Control (PSC) boarding inspection, oil-fingerprinting sample matching, and admiralty tribunal submission.